

# FreshFlow Beverages Pvt. Ltd.

FreshFlow Bottling Plant - Pune Production Facility  
FSSAI: 10012022000345 | ISO 22000:2018 / ISO 14001:2015

## STANDARD OPERATING PROCEDURE

### Quality Inspection Procedure for Water Treatment Line 30

Department: Water Treatment

|                                                  |
|--------------------------------------------------|
| Document Number: SOP-030                         |
| Revision: Rev 3.0                                |
| Classification: CONFIDENTIAL - INTERNAL USE ONLY |
| Prepared by: Engineering / QA Department         |
| Approved by: Plant Manager                       |

## Document Control Information

| Parameter        | Value                                                   |
|------------------|---------------------------------------------------------|
| SOP Number       | SOP-030                                                 |
| Title            | Quality Inspection Procedure for Water Treatment Line 3 |
| Department       | Water Treatment                                         |
| Plant            | FreshFlow Bottling Plant - Pune Production Facility     |
| Prepared By      | Arjun Nair                                              |
| Reviewed By      | Anita Desai                                             |
| Approved By      | Rajesh Kulkarni - Plant Manager                         |
| Issue Date       | 2022-11-26                                              |
| Next Review Date | 2027-06-14                                              |
| Revision         | Rev 2.3                                                 |

## 1. Purpose

Produce water meeting the FreshFlow Beverages water quality specification (TDS <50 ppm, pH 6.5-7.5, microbiologically safe) for use in product production and CIP.

## 2. Scope

Water Treatment operators and Maintenance team.

## 3. Prerequisites and Requirements

### 3.1 Training Requirements

- RO system operation training
- Chemical dosing training

### 3.2 Tools and Materials Required

- Calibrated instruments as listed in the plant calibration schedule.
- Appropriate PPE as specified in Section 4.
- Completed permit-to-work (if required for maintenance activities).
- Access to relevant equipment manual and P&ID drawings.
- CMMS access to create and close work orders.

## 4. Health, Safety and Environmental Requirements

### ! WARNING

This SOP must not be carried out without appropriate PPE and, where required, a valid Permit to Work (SOP-SAF-006). If in doubt about safety, STOP and consult the Safety Officer (EMP009 / EMP017).

#### 4.1 Identified Hazards

- High-pressure RO system (up to 12 bar) - pressure release risk
- UV radiation from UV sterilizer lamps - eye and skin damage risk
- Chemical dosing agents (chlorine, antiscalant, acid) - chemical hazard

#### 4.2 Emergency Response

In case of any emergency during this procedure: Press the nearest Emergency Stop, alert personnel in the vicinity, call the Safety Officer and Shift Supervisor. Refer to the Emergency Response Plan (SOP-SAF-004) and site evacuation procedure.

## 5. Detailed Procedure

#### 5.1 Pre-Task Verification

1. Confirm the relevant equipment is available and no conflicting activities are planned.
2. Check shift handover log for any outstanding issues with this equipment.
3. Verify all required materials, tools, and chemicals are available and correctly labelled.
4. Obtain all required permits (PTW, hot work, confined space as applicable).
5. Brief all involved personnel on this SOP before starting.

#### 5.2 Main Procedure

1. Isolate and LOTO the equipment per SOP-SAF-002. Verify zero energy state.
2. Allow equipment to cool to ambient temperature before touching internal parts.
3. Disassemble relevant sub-assembly according to the equipment manual section 6.
4. Inspect all removed parts. Photograph any damaged or worn components.
5. Clean the housing and contact surfaces. Remove all old gasket material.
6. Install new parts using food-grade lubricants where applicable. Do not over-torque fasteners.
7. Reassemble in reverse order. Verify all fasteners are tightened to specified torques.
8. Remove LOTO and test equipment at low speed/load before returning to production.
9. Verify key parameters (pressure, temperature, vibration) after restart.
10. Record the work in CMMS with parts used, cost, and technician details.

## 6. Quality and Verification Checks

| Verification Check                                   | When              | Responsible | Record          |
|------------------------------------------------------|-------------------|-------------|-----------------|
| All product contact surfaces visually clean          | Before production | Operator    | Shift Log       |
| CIP conductivity $\leq 20$ uS/cm                     | After each CIP    | Operator    | CIP Record      |
| CIP pH 6.5-7.5                                       | After each CIP    | Operator    | CIP Record      |
| Microbiological swab $<10$ CFU/cm <sup>2</sup>       | After each CIP    | QC Lab      | Micro Report    |
| Key process parameters in normal range               | At startup        | Operator    | SCADA Historian |
| First-off product sample - Brix, pH, CO <sub>2</sub> | After startup     | QC Analyst  | QC Lab Report   |

## 7. Non-Conformance and Escalation

If any check fails or if the procedure cannot be completed as written: STOP the task. Do not force equipment to operate outside specification. Escalate to the Shift Supervisor (Suresh Patil / Ravi Deshpande). Raise a Non-Conformance Report (NCR) in the CMMS system. QC Manager (Anita Desai) must be notified for any product safety impact.

## 8. Records and Documentation

| Record Type                 | Frequency                  | Responsible      | Storage Location       |
|-----------------------------|----------------------------|------------------|------------------------|
| Shift Log - Water Treatment | After each shift           | Shift Supervisor | Shift Log Book / SCADA |
| CMMS Work Order             | For every maintenance task | Technician       | FreshFlow CMMS         |
| CIP Record (FFB-QA-CIP-001) | After every CIP cycle      | Operator         | QA Filing System       |
| Instrument Calibration Log  | After calibration          | QC/Maintenance   | Calibration Database   |

## A. Supplementary Notes

---

This SOP should be read in conjunction with the relevant equipment manual and the FreshFlow Beverages Good Manufacturing Practice (GMP) guidelines. Any suggested improvements to this SOP should be submitted to the Engineering Department using the Document Change Request (DCR) form FFB-ENG-DCR-001. This SOP is reviewed annually or whenever a significant process change occurs.

Related SOPs: SOP-SAF-002 (LOTO), SOP-GEN-010 (GMP), SOP-QA-005 (HACCP), SOP-GEN-001 (Shift Handover), SOP-GEN-006 (Plant Startup). For questions on this procedure, contact the Engineering or QA Department.